

Table 1: Node classification results on Movies, Toys, Grocery, ele-fashion, and Reddit-S. Test Accuracy and Test Macro-F1 are percentages reported as mean $\pm$ population standard deviation over seeds 42, 43, and 44. All models use the frozen transductive full-graph protocol, validation-accuracy checkpoint selection, and the fixed task-level Macro-F1 label set.

Model	Movies		Toys		Grocery		ele-fashion		Reddit-S	
	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1
MLP	50.91 $\pm$ 0.15	36.84 $\pm$ 2.22	74.30 $\pm$ 0.29	71.84 $\pm$ 0.38	78.86 $\pm$ 0.48	69.78 $\pm$ 1.18	87.92 $\pm$ 0.04	76.81 $\pm$ 0.31	92.82 $\pm$ 0.01	87.28 $\pm$ 0.03
GCN	53.22 $\pm$ 0.51	45.30 $\pm$ 0.14	78.77 $\pm$ 0.22	75.25 $\pm$ 0.44	81.27 $\pm$ 0.12	71.95 $\pm$ 0.46	85.21 $\pm$ 0.10	73.02 $\pm$ 0.23	93.52 $\pm$ 0.08	88.88 $\pm$ 0.20
GraphSAGE	52.50 $\pm$ 1.93	40.63 $\pm$ 10.84	77.75 $\pm$ 0.31	75.19 $\pm$ 0.22	82.61 $\pm$ 0.23	74.35 $\pm$ 0.42	87.75 $\pm$ 0.26	76.55 $\pm$ 0.93	94.84 $\pm$ 0.05	90.45 $\pm$ 0.17
MMGCN	55.25 $\pm$ 1.00	48.00 $\pm$ 0.75	78.36 $\pm$ 0.25	75.33 $\pm$ 0.43	82.00 $\pm$ 0.11	71.70 $\pm$ 1.31	87.55 $\pm$ 0.05	76.30 $\pm$ 0.30	95.67 $\pm$ 0.15	91.25 $\pm$ 0.39
MGAT	53.59 $\pm$ 0.19	35.94 $\pm$ 1.56	77.84 $\pm$ 0.11	74.05 $\pm$ 0.24	80.94 $\pm$ 0.34	68.40 $\pm$ 1.34	86.88 $\pm$ 0.14	72.35 $\pm$ 0.66	95.70 $\pm$ 0.13	90.56 $\pm$ 0.26
DiP	55.07 $\pm$ 0.18	47.28 $\pm$ 0.56	79.18 $\pm$ 0.36	76.61 $\pm$ 0.26	83.30 $\pm$ 0.26	75.69 $\pm$ 0.47	87.99 $\pm$ 0.28	77.13 $\pm$ 1.31	96.43 $\pm$ 0.05	92.42 $\pm$ 0.38
DGF	51.35 $\pm$ 0.28	31.34 $\pm$ 3.19	78.35 $\pm$ 0.22	74.95 $\pm$ 0.58	81.47 $\pm$ 0.10	69.63 $\pm$ 0.11	86.86 $\pm$ 0.04	72.63 $\pm$ 0.43	95.98 $\pm$ 0.08	90.57 $\pm$ 0.55
DMGC	46.55 $\pm$ 0.57	21.73 $\pm$ 1.56	71.49 $\pm$ 1.83	67.83 $\pm$ 1.44	72.22 $\pm$ 2.86	60.08 $\pm$ 2.84	86.42 $\pm$ 0.25	67.66 $\pm$ 4.52	92.79 $\pm$ 0.31	86.83 $\pm$ 0.32
LGMRec	55.59 $\pm$ 0.30	49.20 $\pm$ 0.09	79.18 $\pm$ 0.02	76.52 $\pm$ 0.13	82.88 $\pm$ 0.08	75.33 $\pm$ 0.21	85.68 $\pm$ 0.31	71.29 $\pm$ 1.52	95.99 $\pm$ 0.04	91.89 $\pm$ 0.13
MoPF	56.19 $\pm$ 0.52	49.49 $\pm$ 0.71	79.64 $\pm$ 0.20	76.99 $\pm$ 0.20	83.42 $\pm$ 0.30	76.18 $\pm$ 0.44	88.03 $\pm$ 0.10	76.99 $\pm$ 0.65	96.37 $\pm$ 0.26	92.11 $\pm$ 0.53

Table 2: Link prediction results on Sports-Copurchase and Cloth-Copurchase. All entries are test metrics reported as percentages. Sports-Copurchase is the formal `unified_sampled_lp_v1` benchmark with seeds 42, 43, and 44; values are mean $\pm$ population standard deviation and checkpoint selection uses validation MRR. Cloth-Copurchase values are included for completeness, but are not a second formal LP benchmark. Validation MRR is omitted.

Model	Sports-Copurchase				Cloth-Copurchase			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
MLP	24.99 $\pm$ 0.35	11.76 $\pm$ 0.31	26.32 $\pm$ 0.34	55.39 $\pm$ 0.70	18.97 $\pm$ 0.20	9.02 $\pm$ 0.15	19.12 $\pm$ 0.21	39.41 $\pm$ 0.46
GCN	32.09 $\pm$ 0.14	17.77 $\pm$ 0.18	35.44 $\pm$ 0.05	64.11 $\pm$ 0.23	20.99 $\pm$ 0.12	11.04 $\pm$ 0.10	21.92 $\pm$ 0.14	41.05 $\pm$ 0.70
GraphSAGE	32.59 $\pm$ 0.24	17.10 $\pm$ 0.16	36.41 $\pm$ 0.26	69.05 $\pm$ 0.41	24.68 $\pm$ 0.13	12.71 $\pm$ 0.10	26.15 $\pm$ 0.12	50.40 $\pm$ 0.22
MMGCN	34.63 $\pm$ 0.30	18.42 $\pm$ 0.33	39.49 $\pm$ 0.26	<u>72.45 $\pm$ 0.30</u>	<u>26.25 $\pm$ 0.04</u>	13.72 $\pm$ 0.03	<u>28.31 $\pm$ 0.10</u>	<u>53.32 $\pm$ 0.17</u>
MGAT	32.61 $\pm$ 0.11	17.17 $\pm$ 0.05	36.29 $\pm$ 0.26	69.11 $\pm$ 0.61	25.22 $\pm$ 0.13	13.16 $\pm$ 0.08	26.93 $\pm$ 0.23	51.26 $\pm$ 0.30
DiP	<u>35.14 $\pm$ 0.21</u>	<u>18.90 $\pm$ 0.39</u>	<u>40.29 $\pm$ 0.36</u>	72.28 $\pm$ 0.52	25.76 $\pm$ 0.82	13.29 $\pm$ 0.62	27.84 $\pm$ 1.03	52.78 $\pm$ 1.35
DGF	34.19 $\pm$ 0.31	18.57 $\pm$ 0.25	38.51 $\pm$ 0.45	70.44 $\pm$ 0.44	25.93 $\pm$ 0.13	<u>13.79 $\pm$ 0.14</u>	27.80 $\pm$ 0.14	51.96 $\pm$ 0.24
DMGC	32.93 $\pm$ 1.77	17.54 $\pm$ 1.57	37.33 $\pm$ 2.43	68.05 $\pm$ 2.33	23.35 $\pm$ 0.11	11.52 $\pm$ 0.07	24.70 $\pm$ 0.08	48.63 $\pm$ 0.24
LGMRec	32.92 $\pm$ 0.03	18.16 $\pm$ 0.08	36.71 $\pm$ 0.14	66.25 $\pm$ 0.09	22.16 $\pm$ 0.20	11.47 $\pm$ 0.22	23.33 $\pm$ 0.28	44.28 $\pm$ 0.29
MoPF	37.48 $\pm$ 0.34	21.56 $\pm$ 0.39	43.01 $\pm$ 0.44	73.04 $\pm$ 0.38	27.91 $\pm$ 0.35	15.43 $\pm$ 0.19	30.49 $\pm$ 0.45	54.57 $\pm$ 0.77